

Taller - Sesión 4 - Series de Tiempo y Python IIE - UNAM

Luis E. Ascencio G.
CIMAT
luis.ascencio@cimat.mx

Abstract Este Notebook incluye una introducción al manejo de Series de Tiempo con Python

```
import numpy as np # Libreria Matematica basica
import pandas as pd # Libreria para manejo, manipulacion y visualizacion de
datos
from pandas import read_excel # funcion para leer archivos de excel

import matplotlib as mpl # Libreria para visualizacion de datos y graficas
import matplotlib.pyplot as plt # Funcion para graficar
import seaborn as sns # Libreria para visualizacion de datos
from pandas.plotting import autocorrelation_plot

from statsmodels.tsa.stattools import acf, pacf
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf

from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.seasonal import seasonal_decompose
from statsmodels.tsa.stattools import adfuller
from statsmodels.tsa.arima_process import ArmaProcess

import session_info
```

```
import sktime
from sktime import datasets
from sktime.utils import plot_series
from sklearn.linear_model import LinearRegression
import statsmodels.api as sm
```

```
from sktime.forecasting.naive import NaiveForecaster
from sktime.forecasting.model_selection import temporal_train_test_split
```

Datos

```
df = pd.read_csv('AirPassengers.csv')
df1 = read_excel('ClayBricks.xls')
df2 = read_excel('Electricity.xls')
df3 = read_excel('MilkProduction.xls')
df4 = read_excel('JapaneseCars.xls')
df5 = read_excel('HouseSales.xls')
```

df1

	Dates	Bricks
0	1956-03-01	189
1	1956-04-01	204
2	1956-05-01	208
3	1956-06-01	197
4	1956-07-01	187
...	...	...
150	1968-09-01	476
151	1968-10-01	443
152	1968-11-01	421
153	1968-12-01	472
154	1969-01-01	494

```
dates=pd.to_datetime(df["Month"])
date1=pd.to_datetime(df1["Dates"])
date2=pd.to_datetime(df2["Month and year"])
```

```
airline=datasets.load_airline()
```

airline

```
Period
1949-01    112.0
1949-02    118.0
1949-03    132.0
1949-04    129.0
1949-05    121.0
```

```

...
1960-08    606.0
1960-09    508.0
1960-10    461.0
1960-11    390.0
1960-12    432.0
Freq: M, Name: Number of airline passengers, Length: 144, dtype: float64

```

```

ts=pd.Series(np.array(df["#Passengers"]),index=dates)
ts1=pd.Series(np.array(df1["Bricks"]),index=date1)
ts2=pd.Series(np.array(df2["Kwh"]),index=date2)
ts3=pd.Series(np.array(df3["Monthly Milk Production per Cow"]))
ts5=pd.Series(np.array(df5["HouseSales"]),index=df5["Month"])

```

```
ts1
```

```

Dates
1956-03-01    189
1956-04-01    204
1956-05-01    208
1956-06-01    197
1956-07-01    187
...
1968-09-01    476
1968-10-01    443
1968-11-01    421
1968-12-01    472
1969-01-01    494
Length: 155, dtype: int64

```

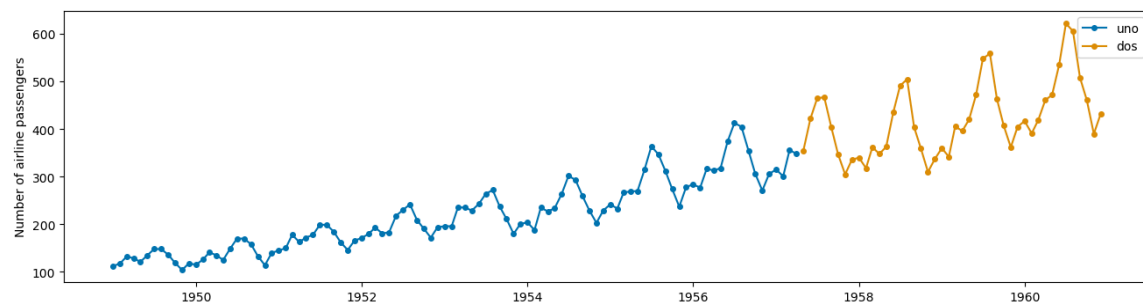
SkTimes Graficador

```
plot_series(airline[:100],airline[100:],labels=["uno","dos"])
```

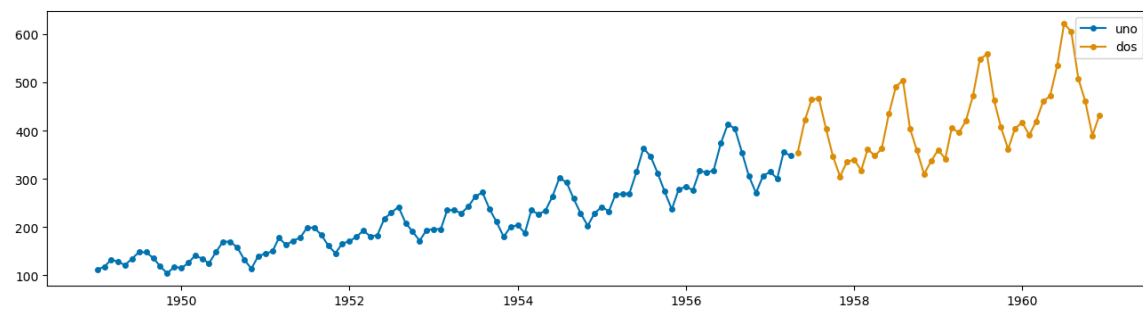
```

(<Figure size 1600x400 with 1 Axes>,
 <Axes: ylabel='Number of airline passengers'>)

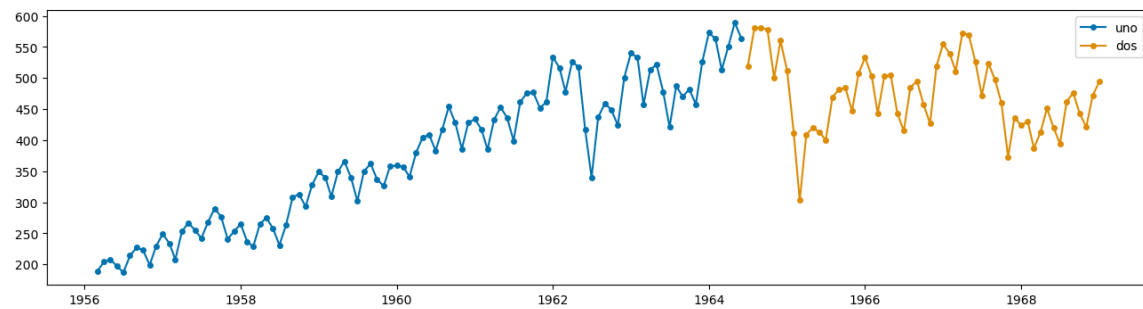
```



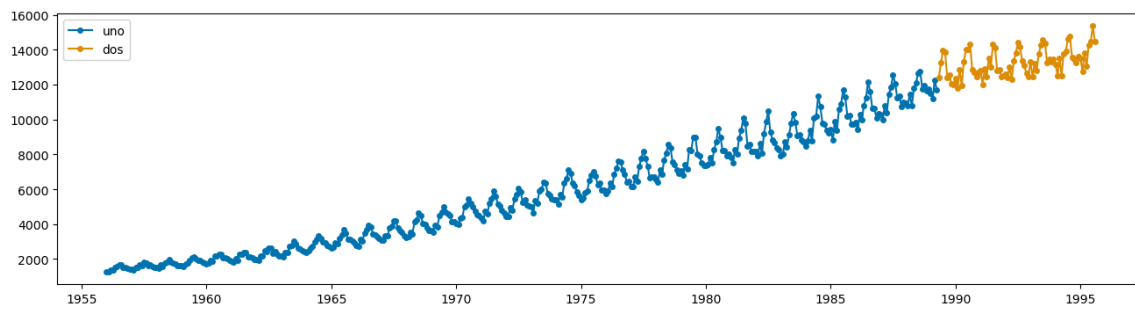
```
plot_series(ts[:100],ts[100:],labels=["uno","dos"])
```



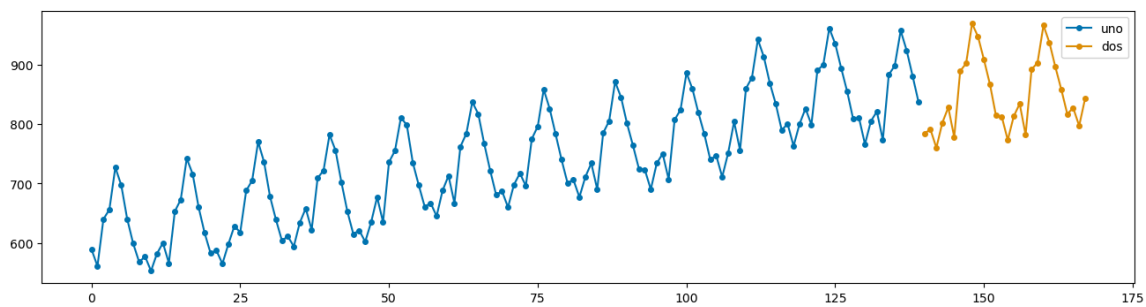
```
plot_series(ts1[:100],ts1[100:],labels=["uno","dos"])
```



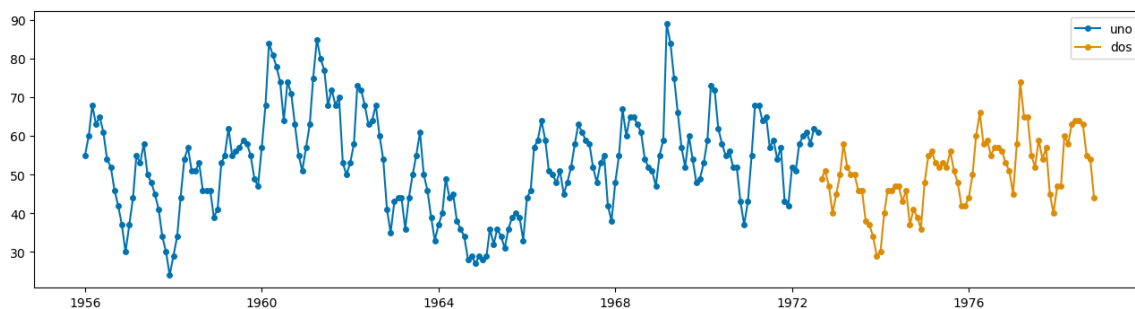
```
plot_series(ts2[:400],ts2[400:],labels=["uno","dos"])
```



```
plot_series(ts3[:140],ts3[140:],labels=["uno","dos"])
```



```
plot_series(ts5[:200],ts5[200:],labels=["uno","dos"])
```



Prediccion Naive sin periodo

Aerolineas

```
AirTrain,AirTest=temporal_train_test_split(airline)
```

```
predic_Air1=NaiveForecaster(strategy="last")
predic_Air2=NaiveForecaster(strategy="mean")
predic_Air3=NaiveForecaster(strategy="drift")
```

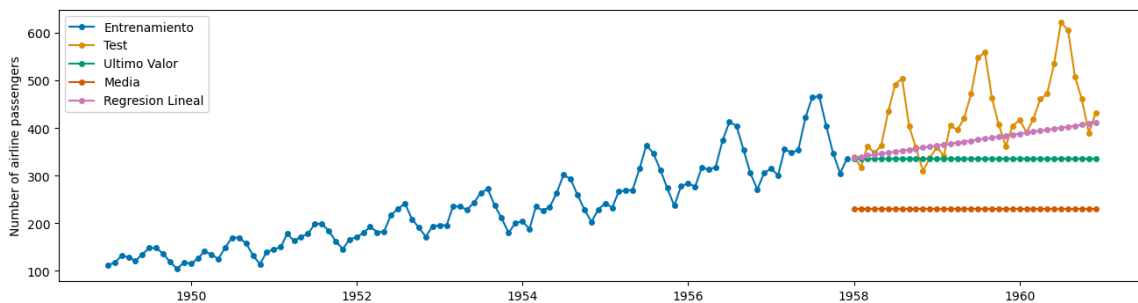
```
predic_Air1.fit(AirTrain)
predic_Air2.fit(AirTrain)
predic_Air3.fit(AirTrain)
```

```
NaiveForecaster(strategy='drift')
```

```
x=np.arange(1,len(AirTest)+1)
y1=predic_Air1.predict(x)
y2=predic_Air2.predict(x)
y3=predic_Air3.predict(x)
```

```
plot_series(AirTrain,AirTest,y1,y2,y3,labels=["Entrenamiento","Test","Ultimo
Valor","Media","Regresion Lineal"])
```

(<Figure size 1600x400 with 1 Axes>,
<Axes: ylabel='Number of airline passengers'>)



Ladrillos

```
ts1.index = ts1.index.to_period(freq="M")
```

```
ts1_Train,ts1_Test=temporal_train_test_split(ts1)
```

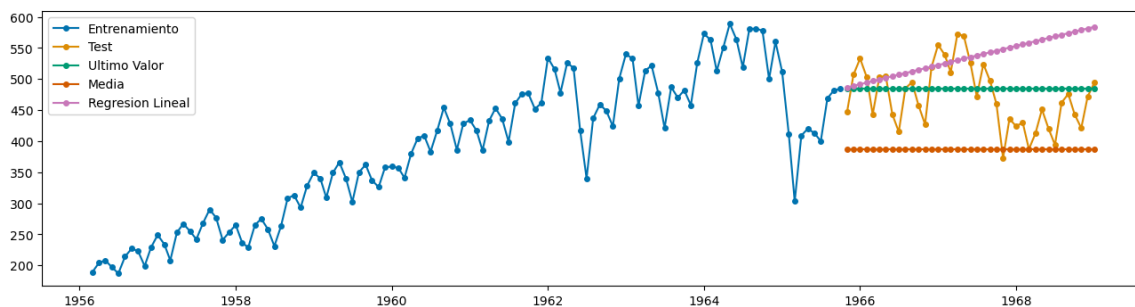
```
predic_ts11=NaiveForecaster(strategy="last")
predic_ts12=NaiveForecaster(strategy="mean")
predic_ts13=NaiveForecaster(strategy="drift")
```

```
predic_ts11.fit(ts1_Train)
predic_ts12.fit(ts1_Train)
predic_ts13.fit(ts1_Train)
```

```
NaiveForecaster(strategy='drift')
```

```
x=np.arange(1,len(ts1_Test)+1)
y1=predic_ts11.predict(x)
y2=predic_ts12.predict(x)
y3=predic_ts13.predict(x)
```

```
plot_series(ts1_Train,ts1_Test,y1,y2,y3,labels=["Entrenamiento","Test","Ultimo
Valor","Media","Regresion Lineal"])
```



Prediccion con Periodos

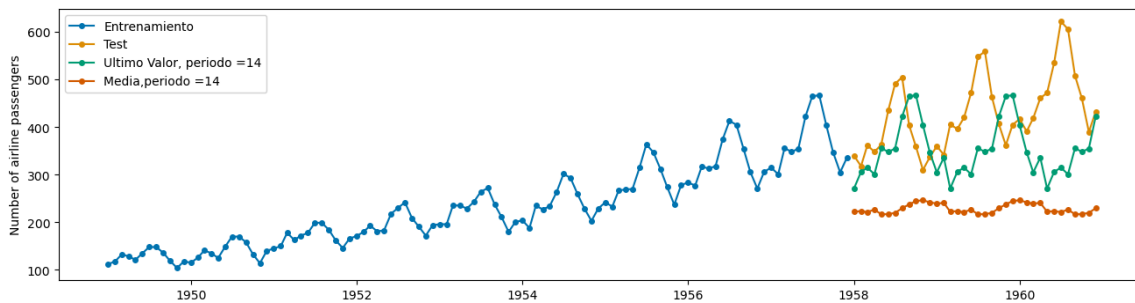
```
predic_Air21=NaiveForecaster(strategy="last", sp=14)
predic_Air22=NaiveForecaster(strategy="mean", sp=14)
```

```
predic_Air21.fit(AirTrain)
predic_Air22.fit(AirTrain)
```

```
x=np.arange(1,len(AirTest)+1)
y21=predic_Air21.predict(x)
y22=predic_Air22.predict(x)
```

```
plot_series(AirTrain,AirTest,y21,y22,labels=["Entrenamiento","Test","Ultimo
Valor, periodo =14","Media,periodo =14"])
```

```
(<Figure size 1600x400 with 1 Axes>,
<Axes: ylabel='Number of airline passengers'>)
```



```

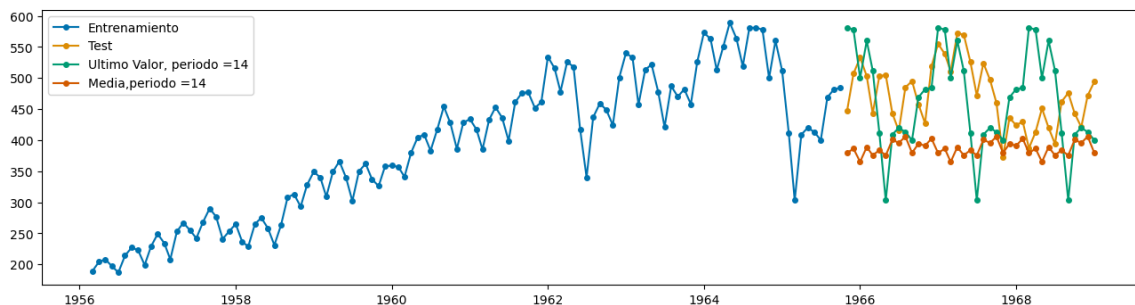
predic_ts1_21=NaiveForecaster(strategy="last", sp=14)
predic_ts1_22=NaiveForecaster(strategy="mean", sp=14)

predic_ts1_21.fit(ts1_Train)
predic_ts1_22.fit(ts1_Train)

x=np.arange(1, len(ts1_Test)+1)
y21=predic_ts1_21.predict(x)
y22=predic_ts1_22.predict(x)

plot_series(ts1_Train, ts1_Test, y21, y22, labels=["Entrenamiento", "Test", "Ultimo
Valor, periodo =14", "Media, periodo =14"])

```



```

from statsforecast import StatsForecast
from statsforecast.models import AutoARIMA
from statsforecast.utils import AirPassengersDF

```

```

/home/leag555/anaconda3/lib/python3.13/site-packages/fs/__init__.py:4:
UserWarning: pkg_resources is deprecated as an API. See https://setuptools.
pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for
removal as early as 2025-11-30. Refrain from using this package or pin to
Setuptools<81.
__import__("pkg_resources").declare_namespace(__name__) # type: ignore

```



```
df.columns=["ds","y"]
```

df

	ds	y
0	1949-01	112
1	1949-02	118
2	1949-03	132
3	1949-04	129
4	1949-05	121
...	...	...
139	1960-08	606
140	1960-09	508
141	1960-10	461
142	1960-11	390
143	1960-12	432

```
df.insert(0, 'unique_id', len(df)*["AirPassengers"])
```

df

	unique_id	ds	y
0	AirPassengers	1949-01	112
1	AirPassengers	1949-02	118
2	AirPassengers	1949-03	132
3	AirPassengers	1949-04	129
4	AirPassengers	1949-05	121
...	...	...	...
139	AirPassengers	1960-08	606
140	AirPassengers	1960-09	508
141	AirPassengers	1960-10	461
142	AirPassengers	1960-11	390

	unique_id	ds	y
143	AirPassengers	1960-12	432

```
sf = StatsForecast(
    models=[AutoARIMA(season_length=12)],
    freq='ME',
)

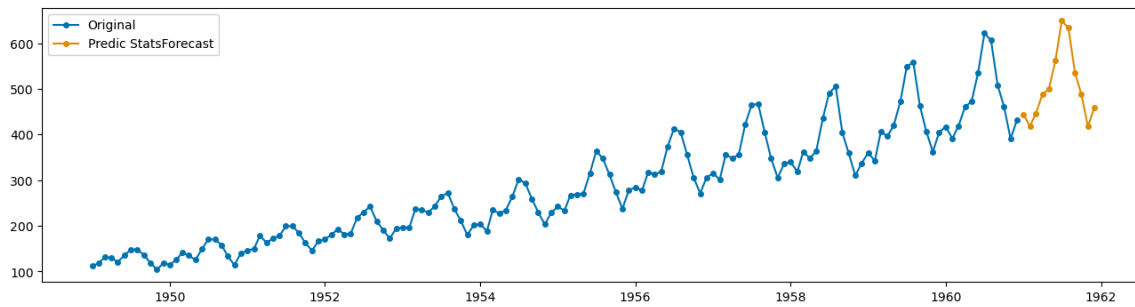
sf.fit(df)
dff_forecast=sf.predict(h=12, level=[95])
```

dff_forecast

	unique_id	ds	AutoARIMA	AutoARIMA-lo-95	AutoARIMA-hi-95
0	AirPassengers	1960-12-31	444.309570	421.279327	467.339813
1	AirPassengers	1961-01-31	418.213745	390.227631	446.199829
2	AirPassengers	1961-02-28	446.243408	412.910034	479.576782
3	AirPassengers	1961-03-31	488.234222	450.621368	525.847107
4	AirPassengers	1961-04-30	499.237061	457.694824	540.779297
5	AirPassengers	1961-05-31	562.236206	517.130981	607.341431
6	AirPassengers	1961-06-30	649.236450	600.822449	697.650513
7	AirPassengers	1961-07-31	633.236389	581.727783	684.744934
8	AirPassengers	1961-08-31	535.236389	480.808289	589.664490
9	AirPassengers	1961-09-30	488.236389	431.037781	545.434998
10	AirPassengers	1961-10-31	417.236389	357.395325	477.077454
11	AirPassengers	1961-11-30	459.236389	396.864746	521.608032

```
date_sf=pd.to_datetime(dff_forecast["ds"])
ts_predic_sf=pd.Series(np.array(dff_forecast["AutoARIMA"]),index=date_sf)
```

```
plot_series(ts,ts_predic_sf,labels=["Original","Predic StatsForecast"])
```



```
session_info.show(html=False)
```

```
-----
matplotlib      3.10.6
numpy            2.3.3
pandas           2.3.3
seaborn          0.13.2
session_info     v1.0.1
sklearn          1.7.2
sktime           0.38.5
statsforecast    2.0.2
statsmodels      0.14.5
-----
IPython          9.6.0
jupyter_client   8.6.3
jupyter_core     5.8.1
jupyterlab       4.4.9
notebook         7.4.7
-----
Python 3.13.5 | packaged by conda-forge | (main, Jun 16 2025, 08:27:50) [GCC
13.3.0]
Linux-6.8.0-85-generic-x86_64-with-glibc2.39
-----
Session information updated at 2025-10-27 07:11
```